

Caltech CS156: Learning From Data

Lecture 02: Is Learning Feasible?

Comprehensive Technical Study Guide · Author: Gemini Notebook · September 2026

1 Introduction & The Core Dilemma of Learning

In Lecture 01, we established the three fundamental criteria for applying Machine Learning:

1. **Existence of a Pattern:** An underlying relationship exists (e.g., creditworthiness based on salary, debt, and residence duration).
2. **No Mathematical Closed-Form:** The pattern cannot be pinned down deterministically via explicit formulas or rules.
3. **Availability of Data:** A historical dataset $\mathcal{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)\}$ is available.

At the conclusion of Lecture 01, a disturbing philosophical paradox emerged: *Can we actually learn an unknown target function $f: \mathcal{X} \rightarrow \mathcal{Y}$ outside the observed sample dataset $\mathcal{D}$?*

1.1 The Impossibility of Deterministic Learning

Strictly speaking, in a purely deterministic sense, **learning an unknown function outside the training set is impossible**. Given any finite dataset $\mathcal{D}$, the unknown target function f could take *any* arbitrary value on unobserved inputs $\mathbf{x} \notin \mathcal{D}$. No matter how well a hypothesis g fits the training data $\mathcal{D}$, f is free to disagree with g completely everywhere else.

1.2 The Probabilistic Solution (PAC Framework)

To make learning feasible, we must shift our objective from **absolute deterministic certainty** to **probabilistic confidence**. We accept that we cannot guarantee $g(\mathbf{x}) = f(\mathbf{x})$ for all unseen inputs, but we can guarantee that g is **Probably Approximately Correct (PAC)** under a probability distribution over the input space.

2 Hoeffding's Bin Model Thought Experiment

To build the mathematical foundation of statistical learning, Professor Yaser Abu-Mostafa introduces an intuitive probability thought experiment: the **Bin Model**.

The Bin Model Setup

Consider an opaque bin containing an infinite number of marbles. Each marble is either **Red** or **Green**.

- Let μ be the unknown fraction of **Red** marbles in the entire bin (**Out-of-Sample** probability).
- We pick a random sample of N marbles independently from the bin.
- Let ν be the observed fraction of **Red** marbles in our sample (**In-Sample** frequency).

2.1 Does ν Tell Us Anything About μ ?

- **Short Answer (Deterministic): NO.** The sample can be 100% green even if the bin contains 90% red marbles. Absolute certainty about unobserved marbles is impossible.
- **Long Answer (Probabilistic): YES.** For a sufficiently large sample size N , the sample frequency ν is **highly likely to be close** to the true bin frequency μ .

3 Hoeffding's Inequality & Properties

The quantitative bridge between the sample frequency ν and the true frequency μ is provided by **Hoeffding's Inequality**.

Hoeffding's Inequality (Single Hypothesis / Single Bin)

For any sample size $N \geq 1$ and any error tolerance threshold $\epsilon > 0$:

$$P(|\nu - \mu| > \epsilon) \leq 2e^{-2\epsilon^2 N} \quad (1)$$

3.1 Key Structural Properties of Hoeffding's Bound

1. **Non-Asymptotic Bound:** Holds for *any* finite integer $N \geq 1$, not just as $N \rightarrow \infty$.
2. **Uniform / Distribution-Free:** The upper bound $2e^{-2\epsilon^2 N}$ depends *only* on N and ϵ . It holds regardless of the true, unknown value of $\mu \in [0, 1]$.
3. **Exponential Decay:** As sample size N grows, the probability of a "bad event" (where $|\nu - \mu| > \epsilon$) decays exponentially fast.
4. **PAC Interpretation:**
 - **Approximately:** The deviation between sample and bin frequencies is within ϵ .
 - **Probably:** The failure risk is bounded by a tiny probability $\delta = 2e^{-2\epsilon^2 N}$.
5. **Logical Inversion (Symmetric Inference):** Causally, the bin ratio μ dictates the distribution of sample ratios ν . However, because $|\nu - \mu| = |\mu - \nu|$, we invert the logic to infer the fixed unknown cause μ from the observed effect ν .

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4 Connecting the Bin Model to Machine Learning

To apply Hoeffding's Inequality to machine learning, we establish an exact formal mapping between the Bin Model components and Machine Learning constructs.

Bin Model Component	Machine Learning Construct
Bin	Input Space $\mathcal{X}$
Individual Marbles	Input vectors $\mathbf{x} \in \mathcal{X}$
Green Marble	Hypothesis agrees with target: $h(\mathbf{x}) = f(\mathbf{x})$
Red Marble	Hypothesis disagrees with target: $h(\mathbf{x}) \neq f(\mathbf{x})$
Bin Ratio μ (Unknown constant)	Out-of-Sample Error $E_{\text{out}}(h) = \mathbb{E}_{\mathbf{x} \sim P} [\mathbb{I}(h(\mathbf{x}) \neq f(\mathbf{x}))]$
Sample Ratio ν (Observed random variable)	In-Sample Error $E_{\text{in}}(h) = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(h(\mathbf{x}_n) \neq f(\mathbf{x}_n))$
Picking N marbles independently	Generating N training examples $\mathbf{x}_1, \dots, \mathbf{x}_N$ i.i.d. from $P(\mathbf{x})$

Table 1: Formal Mapping from the Hoeffding Bin Model to Machine Learning.

4.1 The Essential Stochastic Assumption

We introduce a probability distribution $P(\mathbf{x})$ over the input space $\mathcal{X}$.

- **Requirement:** Training examples $\mathbf{x}_1, \dots, \mathbf{x}_N$ and unseen test examples must be generated **i.i.d.** (**independently and identically distributed**) from the *same* distribution $P(\mathbf{x})$.
- **No Assumptions on f or P :** $P(\mathbf{x})$ and $f(\mathbf{x})$ remain completely unknown.

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5 Verification vs. Genuine Learning

A critical distinction emphasized in Lecture 02 is the difference between **verifying a fixed hypothesis** and **learning a hypothesis from data**.

5.1 Verification (Single Hypothesis h)

If a hypothesis h is chosen **before looking at the training data** $\mathcal{D}$ (like improvising a fixed credit decision rule), Hoeffding's inequality applies directly:

$$P(|E_{\text{in}}(h) - E_{\text{out}}(h)| > \epsilon) \leq 2e^{-2\epsilon^2 N} \quad (2)$$

This guarantees that $E_{\text{in}}(h) \approx E_{\text{out}}(h)$. However, because h was chosen blindly, $E_{\text{in}}(h)$ will likely be very large (poor performance). Verification proves generalization, but does *not* perform learning.

5.2 Genuine Learning (Multiple Hypotheses $\mathcal{H}$)

In actual machine learning, the algorithm $\mathcal{A}$ searches through a hypothesis space $\mathcal{H} = \{h_1, h_2, \dots, h_M\}$ to pick the best hypothesis g that minimizes E_{in} . Because g is selected *after* inspecting $\mathcal{D}$, g is conditioned on the sample data!

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6 The Multiple Bins Dilemma & The Union Bound

When we search through M hypotheses to pick g , simple Hoeffding fails due to **Data Snooping / Selection Bias**.

6.1 The Coin Flipping Analogy

- If 1 person flips a fair coin 5 times, getting 5 heads has probability $(1/2)^5 = 3.125\%$ (rare).
- If 1,000 people each flip a fair coin 5 times, the probability that *at least one person* gets 5 heads is $1 - (1 - 0.03125)^{1000} \approx 99.99999901\%$.
- Reporting the person who got 5 heads as having a "biased coin" is false! It is purely the result of multiple testing across many candidates.

6.2 Mathematical Derivation of the Union Bound

When picking $g \in \{h_1, \dots, h_M\}$ to minimize E_{in} , a "bad event" occurs if $|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon$. This happens if *any* hypothesis $h_m \in \mathcal{H}$ suffers a bad generalization gap:

$$P(|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon) \leq P\left(\bigcup_{m=1}^M (|E_{\text{in}}(h_m) - E_{\text{out}}(h_m)| > \epsilon)\right) \quad (3)$$

$$\leq \sum_{m=1}^M P(|E_{\text{in}}(h_m) - E_{\text{out}}(h_m)| > \epsilon) \quad (\text{Union Bound}) \quad (4)$$

$$\leq \sum_{m=1}^M 2e^{-2\epsilon^2 N} = 2Me^{-2\epsilon^2 N} \quad (5)$$

Generalization Bound for Finite Hypothesis Sets

$$P(|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon) \leq 2Me^{-2\epsilon^2 N} \quad (6)$$

6.3 The Fundamental Trade-off of Machine Learning

The factor $2Me^{-2\epsilon^2 N}$ reveals the core trade-off in machine learning:

- **Small Model Size (M small):** $2Me^{-2\epsilon^2 N}$ is small, guaranteeing $E_{\text{out}} \approx E_{\text{in}}$. However, the model may lack capacity, leading to large E_{in} (**Underfitting**).
- **Large Model Size (M large):** E_{in} can be driven near zero, but $2Me^{-2\epsilon^2 N}$ grows large or trivial (> 1), destroying the guarantee that $E_{\text{out}} \approx E_{\text{in}}$ (**Overfitting**).

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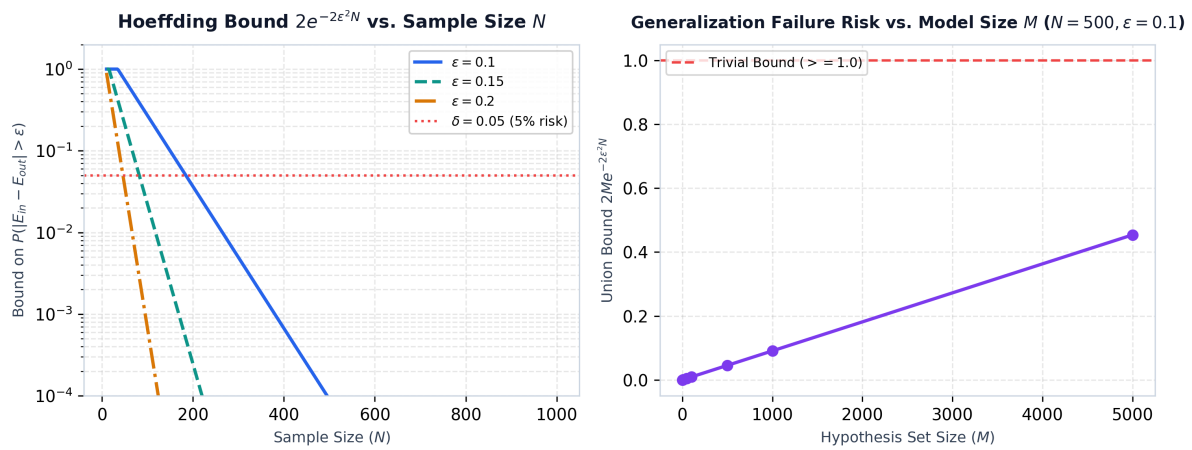


Figure 1: Left: Exponential decay of Hoeffding bound with sample size N across tolerances ϵ . Right: Growth of generalization failure risk with model size M .

7 Visualizing Hoeffding Decay & Complexity Bounds

8 Q&A Highlights & Key Takeaways

Key Technical Insights from Q&A

- Infinite Hypothesis Sets ($M = \infty$):** For models like Perceptrons or Neural Networks, $M = \infty$, rendering the simple Union Bound $2Me^{-2\epsilon^2 N} = \infty$ trivial. *Solution preview:* Replacing M with effective dichotomies $\mathbf{m}_{\mathcal{H}}(N)$ via VC Dimension theory in Lecture 03.
- Cause vs. Effect Inversion:** $P(\mathbf{x})$ generates samples (cause $\rightarrow$ effect), but we infer population parameters μ from sample ν (effect $\rightarrow$ cause). The mathematical symmetry of absolute differences $|\nu - \mu|$ validates this inference.
- Regression Extension:** For real-valued targets, bounded variables use Hoeffding variants based on variance, replacing probability fractions with expected value bounds.